Geometrically Programmed Nano-Resistors for Ultra-Robust Artificial Neural Network Accelerator

by

Giho Lee

B.S. Mechanical Engineering, Seoul National University, 2022
B.S. Electrical and Computer Engineering, Seoul National University, 2022

Submitted to the Department of Mechanical Engineering
in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE IN MECHANICAL ENGINEERING

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

May 2024

© 2024 Giho Lee. All rights reserved.

The author hereby grants to MIT a nonexclusive, worldwide, irrevocable, royalty-free license to exercise any and all rights under copyright, including to reproduce, preserve, distribute and publicly display copies of the thesis, or release the thesis under an open-access license.

Authored by:	Giho Lee Department of Mechanical Engineering April 24, 2024
Certified by:	Jeehwan Kim Associate Professor of Mechanical Engineering, Thesis Supervisor
Accepted by:	Nicolas Hadjiconstantinou Professor of Mechanical Engineering Chair, Department Committee on Graduate Theses

Geometrically Programmed Nano-Resistors for Ultra-Robust Artificial Neural Network Accelerator

by

Giho Lee

Submitted to the Department of Mechanical Engineering
on April 24, 2024 in partial fulfillment of the requirements for the degree of

MASTER OF SCIENCE IN MECHANICAL ENGINEERING

ABSTRACT

Despite the transformative advance in artificial intelligence (AI), the AI processing hardware have not matched the speed and power-efficiency requirement, restricting the realization of the full potential of AI and requiring innovation in AI hardware. Data transmission bottleneck between memory and processor has been pointed out as main source of poor computing speed and power efficiency. By embeding neural weights in hardware to minimize data transmission, non-volatile memory (NVM)-based in-memory computing have expected to have several orders of speed and power-efficiency boosts. However, its practical implementation as a next generation AI hardware has been not successful due to the non-idealities in NVMs including unstability, poor state resolution, challeng in programming, and system-on-a-chip (SoC) incompatibility. This thesis introduces ultra-accurate and ultra-robust geometrically programmed nano-resistor (GPNR) that can overcome NVM non-idealities and enable commercial AI accelerator based on analog in-memory computing. The state-of-the-art 6-bit conductance state resolution and 8-bit stability of nano-resistor was realized by channel geometry optimization and thermodynamically stable material, while SoC imcompatible programming in NVM devices is omited. To evaluate the computing performance, experimental vector-matrix multiplication (NVM) operation were performed, showing 5-bit accuracy operation with 28x28 GPNR array without selectors. Finally, AI inference simulation was performed with simplifed 5x5 cropped MNIST digit image classification task. GPNR-based final classification layer demonstrates 91.0 % accuracy, comparable to the software limit of 93.2 %. The outcomes of this research not only bolster the feasibility of GPNR technology in practical applications but also highlight the potential for future advancements in AI accelerators that can fully harness the capabilities of analog in-memory computing.

Thesis supervisor: Jeehwan Kim
Title: Associate Professor of Mechanical Engineering

Acknowledgments

During my year and a half in Boston, I greatly enjoyed my life as a graduate student and experienced significant growth as a researcher. This growth was facilitated through constructive academic discussions and the invaluable advice from my lab members, as well as the delightful times shared with friends who supported each other in staying energized for our research and studies.

First and foremost, I extend special thanks to Professor Jeehwan Kim for his enthusiastic guidance and support in expanding my ideas into substantial projects. His passion for developing technologies that could change the world has been a major inspiration for me. I am also grateful to Dr. Min-Kyu Song, my closest colleague. He took my initial ideas seriously and helped develop them into robust research topics through numerous discussions and accelerate my research experience with his expert knowledge in micro and nano materials and processes. Additionally, I would like to thank all the members of the Jeehwan Kim group. Dr. Jae Hwan Kim provided invaluable comments that enhanced my understanding of the biomedical field. I also greatly enjoyed the research discussions I had with Dr. Junmin Suh, Dr. Jiho Shin, Dr. Hyunseok Kim, and Doyoon Lee. Outside of research, I had a wonderful time with Jin Feng, Ne Myo Han, Dr. Sangho Lee, Dr. Young Jin Yoo, Dr. Kuangye Lu, Xinyuan Zhang, and others.

Next, I am thankful to the friends in Boston, who made my time in Boston more enjoyable by sharing in activities and frequent interactions. Furthermore, I want to express my gratitude to Professor Yun Seog Lee, Dr. Eunil Kim, Charles Kim, and Dr. Sangwoon Kim for their help in getting me here. Lastly, I sincerely thank my family and friends in South Korea for their unconditional support and love. I deeply wish that one day my research will provide significant benefits to these people.

Contents

Title page 1

Abstract 3

Acknowledgments 5

List of Figures 9

List of Tables 13

1 Introduction 15

1.1 Analog AI accelerator beyond von-Neumann bottleneck 15

1.2 Conventional NVM for analog AI accelerator 18

1.3 NVM Non-idealities and practical analog AI accelerator 19

2 Geometrically Programmed Nano-Resistor (GPNR) 21

2.1 Trade-off between programmability and reliability 21

2.2 Geometrically Programmed Nano-Resistor (GPNR) Concepts 23

3 GPNR Device Development 25

3.1 Device-level analysis 26

3.2 Array-level analysis 27

3.3 Inference-level analysis 30

3.4 GPNR crossbar array fabrication 32

4 GPNR Device Performance and Computing Demonstration 35

4.1 GPNR device performance 36

4.2 Vector-matrix multiplication (VMM) performance 37

4.3 AI inference simulation 38

5 Conclusion 41

References 43

List of Figures

1.1 Conventional von-Neumann architecture and in-memory computing 16
1.2 The schematic of NVM-based analog in-memory computing 17

2.1 The tradeoff between programmability and reliability 22
2.2 The most reliable technology and the most reliable resistive component . . . 23
2.3 The GPNR concept and its potential in analog in-memory computing (IMC). (a) The GPNR concept. The geometry of the channel determines the conductance value representing neural weight in AI. (b) Benchmarking with other in-memory computing platforms and conventional processors 24

3.1 Technical considerations for GPNR-based analog in-memory computing implementation 25
3.2 The parameters determining the GPNR device conductance 26
3.3 The GPNR parameter analysis. (a) Normalized absolute conductance gradient map. (b) Normalized logscale area map. (c) Allowed pairs of length and width. Regions a,b, and c are inappropriate for written reasons. 27
3.4 Circuit level non-idealities regarding undesired resistance 28

3.5 Required GPNR conductance analysis based on circuit simulation. (a) Maximum allowed GPNR conductance depending on the finite input and output resistance. This simulation assumed 8-bit GPNR conductance and operation accuracy with interconnect conductance of 4.9 S. (b) Maximum allowed conductance ratio between GPNR device and interconnect. This simulation assumed 10 mS of IO terminal conductance and 4.9 S of interconnect conductance. 29
3.6 The schematic of weight matrix representation of the GPNR array 31
3.7 AI inference simulation with different weight encoding strategies. (a) The inference accuracy for 25-31-31-10 artificial neural networks for 5x5 simplified MNIST handwritten digit classification task. (b) The inference accuracy for 63-63-63-10 artificial neural networks for 5x5 simplified MNIST handwritten digit classification task. 31
3.8 Automatically drawn 32x32 GPNR array resistor pattern 33
3.9 ITO channel-based GPNR fabrication process 34

4.1 Fully-fabricated GPNR device crossbar array. Scalebar lengths are 1 mm and 100 um. 35
4.2 The GPNR device performance. (a) Normalized conductance states of 80 GPNR devices having five groups of 0th-15th conductance states. Conductance states are normalized by twice each group's average conductance value. (b) Normalized conductance states of 784 GPNR devices measured for 24 hours. (c) Normalized conductance error of 784 GPNR devices measured for 24 hours. The inset figure in (c) is drawn with the same data by rescaling the y-axis. 36

4.3 The GPNR device vector-matrix multiplication (VMM) performance. The experimental VMM operation results of (a) 8x8, (b) 16x16, and (c) 28x28 GPNR arrays with randomly generated input vectors ranging from 0 to 1. The VMM outputs are compared with estimated VMM results calculated with a measured conductance array. 37

4.4 Fully-fabricated neural weight matrix. From left to right, the statistics of the first, second, and third neural layers are summarized. * The number of distinguishable of the first layer was calculated without first column values. . 38

List of Tables

3.1 The AI inference simulation configuration 32

4.1 5x5 simplified MNIST handwritten image classification task inference simulation results 39

Chapter 1

Introduction

This chapter is devoted to introducing the concept of analog artificial intelligence (AI) accelerator and the limitation of the state-of-the-art technology. In the first section, the critical bottleneck in conventional computing architecture limiting AI computing performance and the role of analog AI accelerators in solving this problem will be discussed. The second section will illustrate the state-of-the-art non-volatile memory (NVM) device components utilized for analog AI accelerators, while the final section will explain the remaining challenges preventing market-ready analog AI accelerators.

1.1 Analog AI accelerator beyond von-Neumann bottleneck

Artificial intelligence (AI) has significantly advanced in the last two decades in areas such as computer vision [1]–[4], natural language processing [5], [6], and robotics [7], [8]. Recently, AI performance has been aiming at human-comparable intelligence, demonstrating astonishing context understanding ability, general functionality, and information generation fueled by generative AI and foundation model [6], [9].

In spite of this significant potential to improve human lives, the implementation of AI-

based technology is challenging due to its resource-hungry nature. The exponential growth of AI computing performance due to data-intensive training and increasing neural network sizes is no longer sustainable as computing hardware struggles to accommodate this growth. For instance, the size of the state-of-the-art large language model based on a transformer has doubled its size every two months while the 'Moore's law' stating that the number of transistors integrated into a chip is doubled every two years is failing [10], [11]. At the same time, the massive power consumption for AI training and inference also limits AI applications in our daily lives due to its expensive cost and mobile incompatibility while being comparable to one nation's energy demand [12]. These unmet computing performance and unsustainable power consumption necessitate new computing hardware with unprecedented computing capacity and power efficiency.

To reinvent the processor, the critical limitation of the conventional computer has been widely discussed, and the von Neumann bottleneck has been pointed out as a key problem (Fig. 1.1). The von Neumann bottleneck states the nature of the von Neumann computing architecture, the most common computing architecture, requires the data to move across memory and processor every time processing and storing data are performed, and this may result in a data bottleneck and subsequential limited computing speed and undesired power consumption. This bottleneck significantly limits computing performance in data-intensive AI.

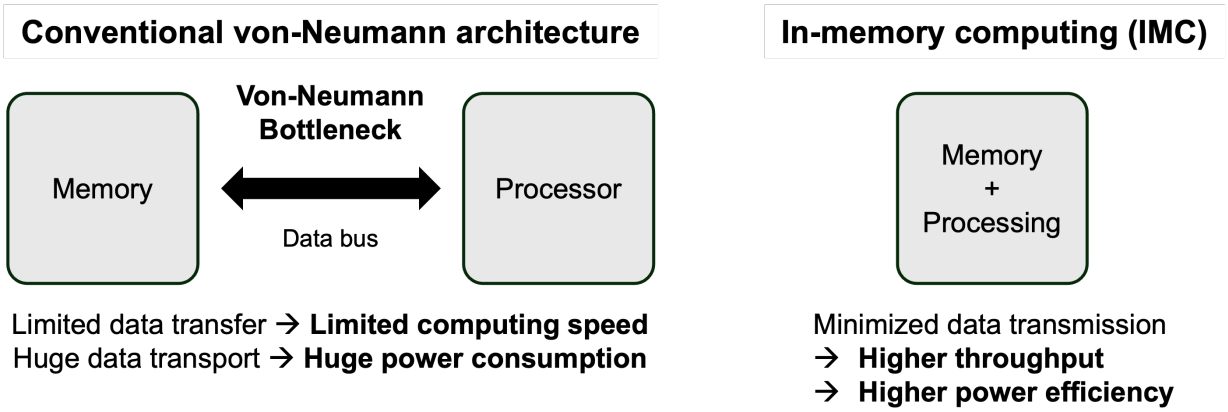


Figure 1.1: Conventional von-Neumann architecture and in-memory computing

The researchers have investigated to solve this problem, and one of the most promising ways is in-memory computing in which the data storing and processing happens in the same unit and the data movement is minimized (Fig. 1.1) [13]. Moreover, non-volatile memory (NVM)-based analog in-memory computing has been introduced, demonstrating its potential to provide several orders of speed and power efficiency boosts over conventional AI processors based on its maximized parallel computation and analog information encoding [14].

The simplest structure of an NVM-based analog in-memory computing platform for vector-matrix multiplication (VMM), a core AI computing operation, is illustrated in Fig 1.2. The crossbar array with non-volatile memories on each cross point stores the neural weight matrix information with their conductance values. When the input vector is given as analog voltages to each of the columns, the weighted sums of input vectors are collected into each column as analog current values corresponding to the result of the VMM operation. Theoretically, this operation can be performed in constant time regardless of the number of bits, inputs, and outputs, which provides enormous speed acceleration. For this reason, this platform is also referred to as an analog AI accelerator.

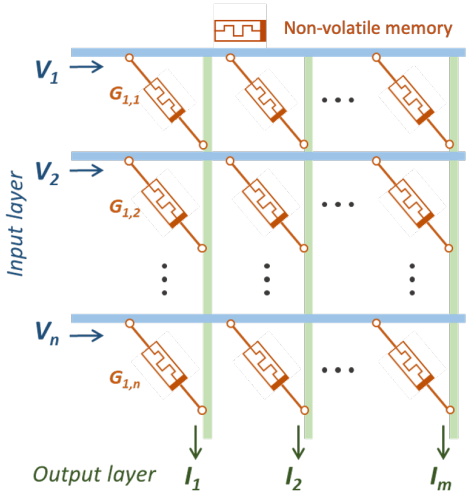


Figure 1.2: The schematic of NVM-based analog in-memory computing

1.2 Conventional NVM for analog AI accelerator

Non-volatile memories (NVMs) are the essential components in analog AI accelerators, storing neural weights of AI architecture and critically affecting computing performance. To support various VMM operations, NVMs are designed to have tunable conductance states, and different kinds of conductance programming mechanism are adopted to demonstrate various NVMs including resistive random-access memory (RRAM), phase-change random-access memory (PCRAM), ferroelectric tunnel junction (FTJ), and magnetic tunneling junction (MTJ) [15]–[17].

Resistive random-access memories (RRAMs) are usually made with the channel of metal oxides [18]–[20]. The first kind of RRAMs utilizes active metals as electrodes so that they can dynamically diffuse into the channel under an electrical field and change the conductance of the channel. In another kind of RRAMs, oxygen ion vacancy moves under an electrical field, and the remaining metallic atoms form conductive paths, resulting in conductance change.

Phase-change random-access memories (PCRAMs) are based on chalcogen-rich alloys providing reversible phase transition [21]. When a high current is applied, the channel melts by resistive heating, and a passive cooling process is followed, changing the portion of the channel from crystalline to an amorphous phase, the low conductance state. Then, relatively small resistive heating is applied to gradually grow the crystalline regime and increase the conductance value. Due to this mechanism, PCRAMs only allow unipolar switching.

Ferroelectric tunnel junction (FTJ) is based on a nano-meters thick ferroelectric channel containing crystalline unit cells [22]. In those cells, the orientation of the transmission coefficient can be adjusted by external bias; thereby, the out-of-plane conductance can be programmed. However, FTJs have a finite potential for analog in-memory computing because of their poor endurance, retention, and the limited number of distinguishable states.

Finally, magnetic tunneling junction (MTJ) has two ferromagnetic layers isolated by a thin insulating layer [23]. One ferromagnetic layer can be switched, while the other ferro-

magnetic layer acts as a reference without changing its magnetic polarization. When both channels have the same polarization, the device is ON state and has a high conductance value, and vice versa for the antipolarization. Unfortunately, MTJs only represent on-and-off states, making this non-optimal for analog in-memory computing.

Among NVMs introduced in the previous section, resistive random-access memory (RRAM) and phase-change random-access memory (PCRAM) demonstrate superior properties for the practical analog AI accelerator implementation [17]. To be specific, they can represent a relatively large number of analog states. Rao et al. recently demonstrated thousands of distinguishable states in RRAM by introducing a novel denoising technique [24]. In addition, endurance and retention are other benefits of RRAMs and PCRAMs over other types of NVMs [17]. For this reason, the development of system-level analog AI accelerators has been largely based on RRAMs and PCRAMs.

1.3 NVM Non-idealities and practical analog AI accelerator

In the last few years, the system-level analog in-memory computing based on state-of-the-art RRAMs and PCRAMs has been actively demonstrated, justifying its compelling role in future AI processor [25]–[29]. In these demonstrations, they not only fabricated the NVM crossbar array but also integrated peripheral circuits such as voltage driving, current measurement, and data handling circuits. Therefore, with this chip-level integration, they were able to experimentally perform AI inference tasks with deep neural networks.

However, when the in-memory computing solely relied on the NVM devices, the inference performance was limited due to the device non-idealities like the limited number of states, state stochasticity, and drifting state value [26]–[30]. In other words, the NVM's non-idealities result in the degenerated performance of analog value representation and subsequential penalty in AI computing results. To overcome this problem, NVM-based in-

memory computing cores are cointegrated with static random-access memory (SRAM)-based in-memory computing cores [25]. With the aid of SRAMs, this practice was able to accomplish the smallest AI inference performance degeneration and illustrate the most practical and compelling solution in the current technology. Nevertheless, the device with robust but precise conductance states will be the ideal solution that can realize the most efficient and fast way of in-memory computing, even though it can be further integrated with SRAMs for advanced functionality.

Another critical bottleneck of current NVM devices for practical AI accelerator applications is conductance state programming. Most programming methods developed for the highest state programming accuracy are based on time- and power-consuming feedback loop algorithms [24], [27]. Moreover, the high current and voltage levels for the programming require a large peripheral circuit area and would not fit into the system-on-chip operating condition. These additional burdens in programming have hampered the implementation of the NVM-based AI computing core into commercial processors.

In summary, in spite of the significant potential of NVM-based AI in-memory computing for high throughput and efficient AI computing, the drawbacks of NVMs have circumscribed its realization. Therefore, the resistive component with a robust and accurate conductance state programmed by a simple method could overcome this bottleneck and pave the way for practical analog in-memory computing.

In the remaining part of this thesis, a new resistive device, a geometrically programmed nano-resistor (GPNR), is suggested with the potential to be the resistive component. In Chapter 2, GPNR is introduced, along with the rationale for how this concept can be more robust and accurate. Then, Chapter 3 discusses practical challenges in realizing the GPNR concept and how those are addressed. Chapter 4 introduces its preliminary data, showing significant robustness and accuracy of the conductance state. Finally, Chapter 5 concludes with the interpretations of the results and future impact.

Chapter 2

Geometrically Programmed Nano-Resistor (GPNR)

In this chapter, the new concept called geometrically programmed nano-resistor (GPNR) is introduced with the several reasons why this device can address the critical challenge in NVM-based analog in-memory computing for realizing the practical NVM-based analog AI accelerator.

2.1 Trade-off between programmability and reliability

Given that the poor reliability of NVMs was the pivotal limitation to implementing NVM-based analog AI accelerator matching industry-level reliable computing, the origin of the poor reliability must be visited. Of course, the primary cause of changing conductance states is the unstable or active properties of the NVM channels, and this is imperative for tunable channels providing programmability to NVM devices as summarized in Fig. 2.1.

To elaborate more in detail, the device with high tunability should have a low energy barrier requiring small programming energy and providing high-frequency switching. Inevitably, the low energy barrier tolerates the device from drifting to thermodynamically stable conditions, immediately bringing forth a problem of poor conductance state retention.

Furthermore, high stochasticity in the short-time domain is followed by a low energy barrier due to readily reversible channel condition if the channel is carefully treated with special method [24]. Therefore, the first assumption of NVM devices was always programmability, accompanying the problem of degenerated reliability of NVMs and AI computing operations.

However, for NVMs to be practically useful in analog in-memory computing, the reliability of computing results should be minimally compromised. Although this limited reliability can be compensated by using additional static random-access memory (SRAM) when additional power, time, and area for the extra circuitry is allowed [25], there are numerous applications having a limited power budget, low-latency requirement, and minimal chip size requirement (e.g., edge AI computing, on-body bio-signal processing, and trading applications). In these applications, the AI accelerator should maximally utilize NVM-dominant analog in-memory computing, requiring high NVM reliability. Moreover, if the NVM has ultra-stable performance and additional resources are also provided, the AI processor can push the limit of AI acceleration by mostly relying on the NVMs rather than SRAM.

This thesis comes from the next question: Does solving the reliability problem over programmability provide the practical analog in-memory computing solution? If we can realize the most reliable resistive component, we can first directly deal with applications requiring reliable, ultra-fast, and predetermined functionality. Indeed, many of the AI technologies

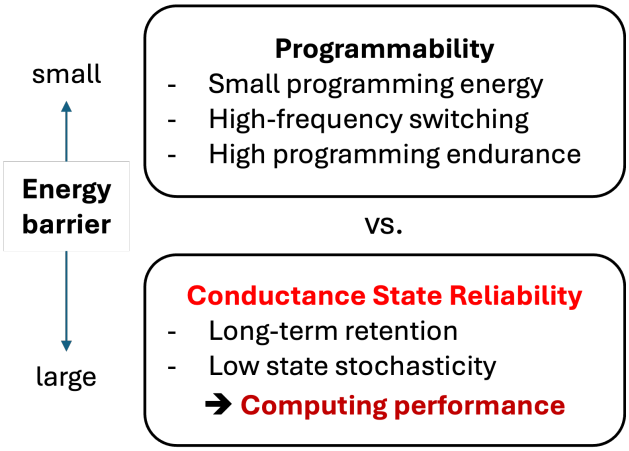


Figure 2.1: The tradeoff between programmability and reliability

in our daily lives are well-defined problems, and they can be the first candidates which the most reliable resistive component can benefit.

2.2 Geometrically Programmed Nano-Resistor (GPNR) Concepts

Given this observation, this thesis suggests geometrically programmed nano-resistor (GPNR) as the most reliable resistive component that could tackle the previous bottleneck of NVMs. GPNR device has a thermodynamically stable channel, leaving minimal room for conductance state change (Fig. 2.2). Furthermore, its conductance state is programmed in channel geometry by nano-precision photolithography technology, which is the most advanced technique, specially optimized for cycle-to-cycle reliability and accuracy in human history. In this way, the GPNR device could provide substantially improved conductance state stability and accuracy, as well as no extra burden of programming power, time, and circuitry.

One example of the structural design of the GPNR device is illustrated in Fig. 2.3a. The conductance value of the channel is governed by the channel material conductivity and the channel trace geometry, such as width, length, thickness, and the number of corners. Those geometry parameters are carefully configured to create specific conductance values corresponding to the neural weight in deep neural networks. In addition, the channel material conductivity is adjusted to determine the range of conductance states. Two electrode lines isolated from each other are making contact with resistive channel traces so that the GPNR

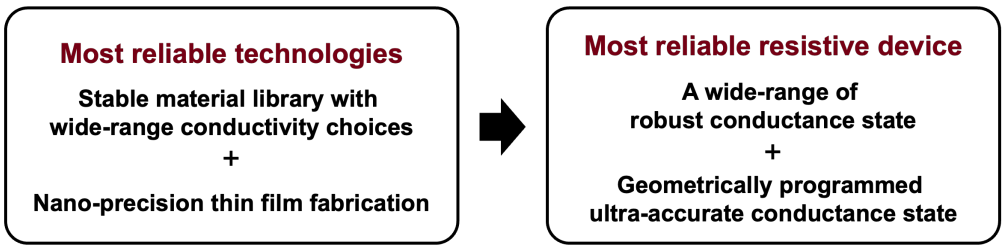


Figure 2.2: The most reliable technology and the most reliable resistive component

device can work as a two-terminal device. Then, this GPNR device can be located on the cross points of the crossbar array shown in Fig. 1.2 to construct a main component of the analog in-memory processor.

By utilizing the most reliable technologies to realize the most reliable resistive component, the GPNR, the most reliable in-memory computing with powerful computing performance, can be implemented (Fig. 2.3b). Since the GPNR-based in-memory computing also utilizes parallel analog computing in NVM-based in-memory computing, it has at least two orders of improvement in AI computing performance compared to conventional graphic processing units (GPUs) and central processing units (CPUs) [10]. In addition, compared to the fusion implementation of NVM- and SRAM-based in-memory computing [25], if analog in-memory computing mainly relies on the GPNR platfrom, the large number of conductance states in GPNR devices will provide additional speed and power benefits. Furthermore, GPNR-based analog in-memory computing does not require power-consuming programming circuitry, resulting in slightly higher operation efficiency as well as at least three orders longer AI computing reliability than NVM-based in-memory computing. Note that the GPNR is not reconfigurable after the fabrication. It would, therefore, have restricted general computing performance. In summary, GPNR-based analog in-memory computing has the potential to realize the most reliable, efficient, and fast AI acceleration for predetermined AI applications.

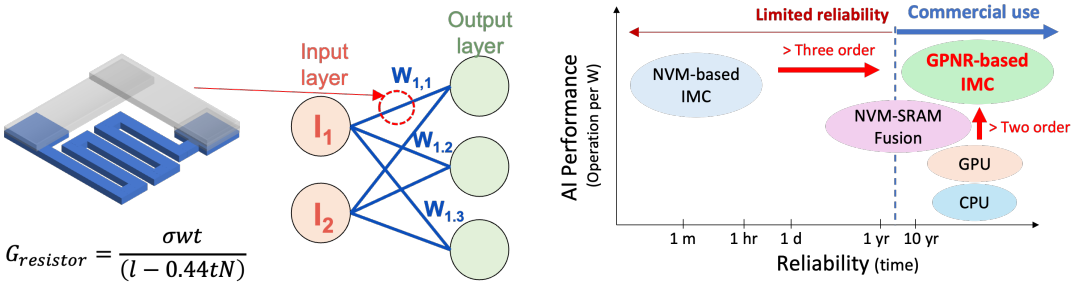


Figure 2.3: The GPNR concept and its potential in analog in-memory computing (IMC). (a) The GPNR concept. The geometry of the channel determines the conductance value representing neural weight in AI. (b) Benchmarking with other in-memory computing platforms and conventional processors

Chapter 3

GPNR Device Development

In this chapter, the critical considerations for robust, accurate, and efficient GPNR-based analog in-memory computing are discussed, summarized in Fig. 3.1. The discussion proceeds from bottom to top. Starting with device geometry design, circuit-level consideration, the required conductance ratio between interconnect and GPNR devices, is followed. Then, the system is further optimized for high-accuracy AI computing in terms of conductance to weight mapping strategy. The final chapter introduces automatic mask drawing software and how the GPNR array is fabricated.

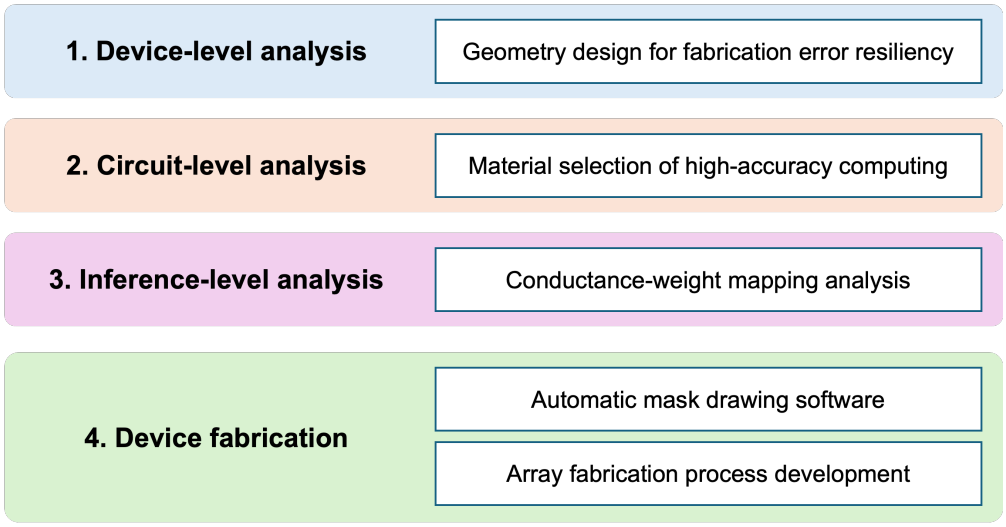


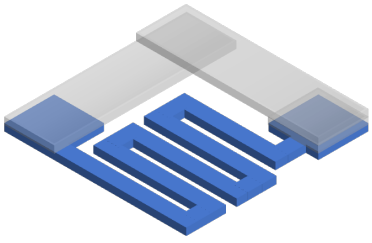
Figure 3.1: Technical considerations for GPNR-based analog in-memory computing implementation

3.1 Device-level analysis

Even if stable channel material provides a robust conductance state, the accurate programming of conductance states is imperative for reliable GPNR-based in-memory computing performance. For this reason, the geometry should be first fabricated with utmost precision. However, even nano-precision photolithography technology cannot realize faultless dimensions. Therefore, the geometry of the channel trace should be carefully designed to have tolerance for minor dimension errors.

Among the parameters determining GPNR conductance values, summarized in Fig. 3.2, channel conductivity and trace thickness do not impact on the weight value accuracy since these parameters impact all GPNR devices on the chip by scaling all of their conductance values, which means relative value of conductance and weight accuracy will not be changed depending on this parameters. Additionally, the number of corners is the parameter dependent on the cell size, trace width, and trace length. Thus, trace width and length are the critical parameters for evaluating the dimension error tolerance of GPNR geometry.

Since the conductance value of the resistor is reciprocally proportional to trace length, it is important to have a long enough trace. This is verified by the calculation shown in Fig. 3.3a as the conductance change gets larger when the trace length is shorter. However, there are other factors of consideration. First, for effective device integration, each GPNR device should be located within the unit cell. Since the area footprint is defined by trace



$$G_{resistor} = \frac{\sigma w t}{(l - 0.44tN)}$$

- σ Channel material conductivity
- w Trace width
- l Trace length
- t Trace thickness
- N Number of corners

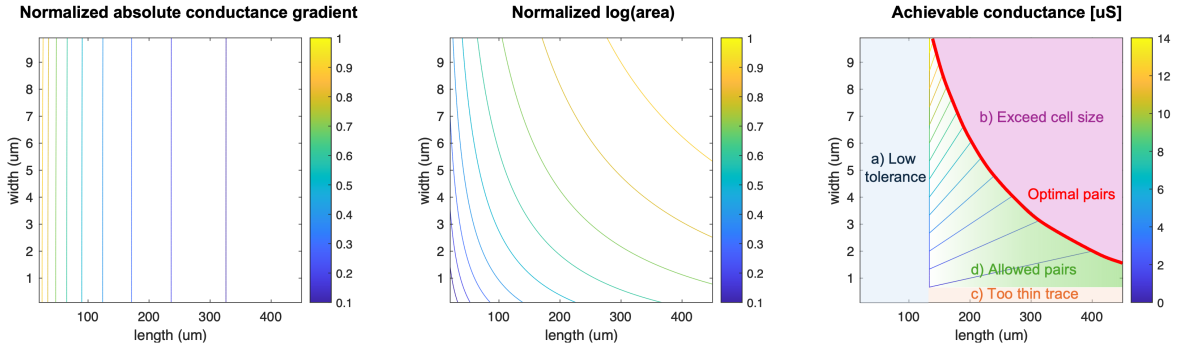
Figure 3.2: The parameters determining the GPNR device conductance

width and length pairs as illustrated in Fig. 3.3b, the desired pairs should have a smaller footprint than the unit cell size. Furthermore, as photolithography has a minimum achievable critical dimension, the trace width should be larger than the minimum critical dimension.

By combining all of these considerations, desired pairs should satisfy the following conditions: a) high enough tolerance for dimension error, b) area footprint smaller than the unit cell size, and c) trace width wider than the critical dimension of utilized photolithography. Shown in Fig. 3.3c, the pairs inside the green allowed region are allowed to be used to accomplish a certain degree of dimension tolerance. Meanwhile, as the dimension tolerance gets larger when the trace length gets longer, the pairs on the curve line at the interface of the allowed region and area-restricted region are selected for the utmost GPNR geometry programming accuracy in this thesis.

3.2 Array-level analysis

Besides the reliability of the GPNR device, array-level non-idealities are other crucial factors affecting the computation performance of resistive component-based analog in-memory computing. The most simple but non-realistic assumption in the ideal VMM operation shown in Fig. 1.2 is negligible resistance of the interconnect. In a practical resistive component



(a) (b) (c)

Figure 3.3: The GPNR parameter analysis. (a) Normalized absolute conductance gradient map. (b) Normalized logscale area map. (c) Allowed pairs of length and width. Regions a,b, and c are inappropriate for written reasons.

crossbar array, there are two undesired finite conductance components: 1) input and output (IO) terminals (purple resistors in Fig. 3.4) and 2) pieces of interconnect (red resistors in Fig. 3.4). If those parts have finite conductance values, the input voltage from input terminals experiences a voltage drop at the input terminal and along the lateral interconnect line. Similarly, the column interconnect lines are supposed to be grounded, which can not be true if the output resistance and interconnect resistance are not zero. Therefore, addressing these non-idealities is vital for the computing resolution matching the GPNR device conductance resolution.

In those cases, the only way to solve this problem is to ensure the largest conductance ratio between GPNR device conductance and undesired finite conductance, thereby making undesired finite conductance negligible. Since there is a limitation in increasing the conductance of the IO terminal and interconnect, the GPNR device conductance should allow large room for engineering with an appropriate selection of channel material with specific conductivity. Therefore, circuit simulations for the required conductance ratio are performed (Fig. 3.5), and the material selection is performed on this information.

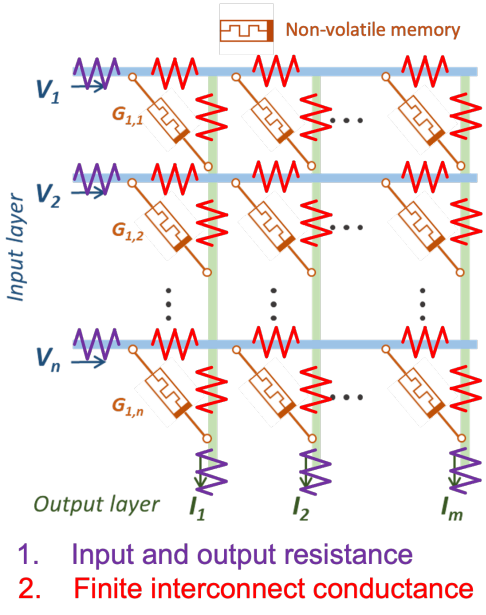
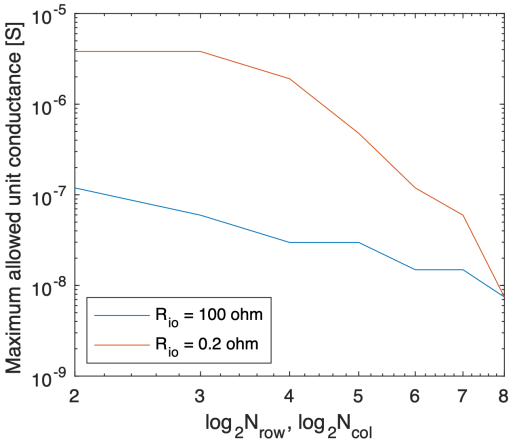


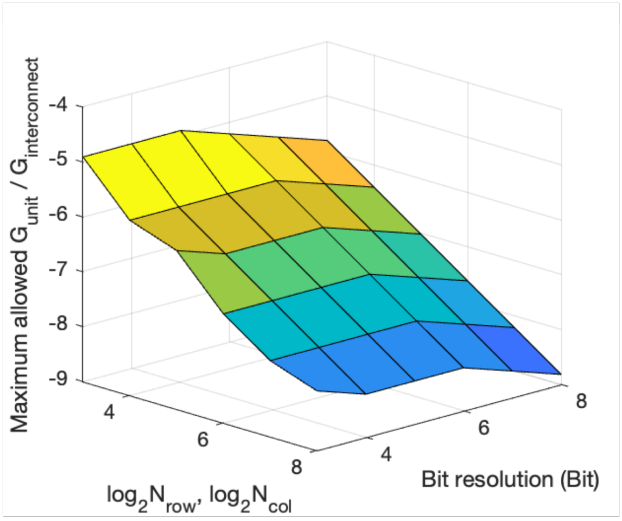
Figure 3.4: Circuit level non-idealities regarding undesired resistance

As shown in Fig. 3.5a, the GPNR device unit conductance, corresponding to the conductance difference between nearest conductance states, should have a larger value for non-deteriorated computing resolution when the non-ideal IO resistance gets larger. Interestingly, although the required unit conductance gets smaller as the number of input and output increases in both IO resistance values, the gap between the two cases is reduced following the increase in input and output numbers and even becomes negligible when the input and output numbers are 256. This implies that generally, having small IO resistance is beneficial for securing a large room for material conductivity, but with a large number of IO channels, the impact of IO resistance would be negligible compared to the voltage drop across the interconnect line.

Given the fact that the bit resolution and the IO channel number can be different depending on the application, the impact of those parameters on the maximum allowed conductance ratio between unit conductance and interconnect conductance was simulated (Fig. 3.5b).



(a)



(b)

Figure 3.5: Required GPNR conductance analysis based on circuit simulation. (a) Maximum allowed GPNR conductance depending on the finite input and output resistance. This simulation assumed 8-bit GPNR conductance and operation accuracy with interconnect conductance of 4.9 S. (b) Maximum allowed conductance ratio between GPNR device and interconnect. This simulation assumed 10 mS of IO terminal conductance and 4.9 S of interconnect conductance.

Generally, the larger number of IO channels and the larger bit resolution computing require the smaller $G_{unit}/G_{interconnect}$ value. The data points were well fitted with the plane governed by equation 3.1,

$$\log_{10} G_{unit}/G_{interconnect} = -0.1390 \times \log_2 N_{row,col} - 0.6178 \times (N_{Bit\ resolution}) - 2.6567 \quad (3.1)$$

where G_{unit}, $G_{interconnect}$, $N_{Bit\ resolution}$, and $N_{row,col}$ represent unit conductance, interconnect conductance between two cross points, bit resolution, and the number of rows and columns, corresponding to IO channels.

Based on this analysis, the maximum allowed conductance ratio for 32x32 GPNR array of 4-bit, 6-bit, and 8-bit resolution are 7.62×10^{-6}, 4.77×10^{-7}, and 2.98×10^{-8} respectively. Given that the channel trace thickness, width, and length for the state 1 geometry of 4-bit GPNR devices are 100 nm, 2 µm, and 400 µm, the appropriate conductivity range for 4-8-bit resolution and interconnect conductance of 1 S is from 15200 to 954 S/m.

3.3 Inference-level analysis

To maximize the AI computing performance of GPNR-based analog in-memory computing, efficient weight representation on the GPNR array and effective device development strategy is of importance.

To begin with, as discussed in the introduction, data loading requires a significant amount of time, and it is critical to exploit maximum computation capacity from the limited size of the GPNR array. Meanwhile, the conductance of the GPNR device is always positive in nature, and they need reference conductance to represent the minus weight value, requiring some portion of the array to be assigned to reference conductance. For this reason, the GPNR array is designed to have reference conductance in the last column, which maximizes the weight matrix representation within the same space (Fig. 3.6).

Furthermore, even if the maximum achievable resolution of GPNR devices is not changed, there still remains an unanswered engineering question: Which is the better direction between engineering to accomplish higher accuracy for quantized conductance values and using analog conductance values, although it is not fully optimized? To answer this question, the AI inference simulation with different strategies was conducted where the result is summarized in Fig 3.7, and the configurations are listed in Table 3.1.

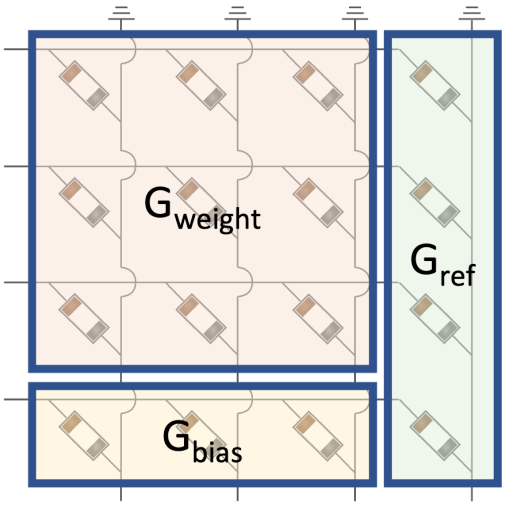
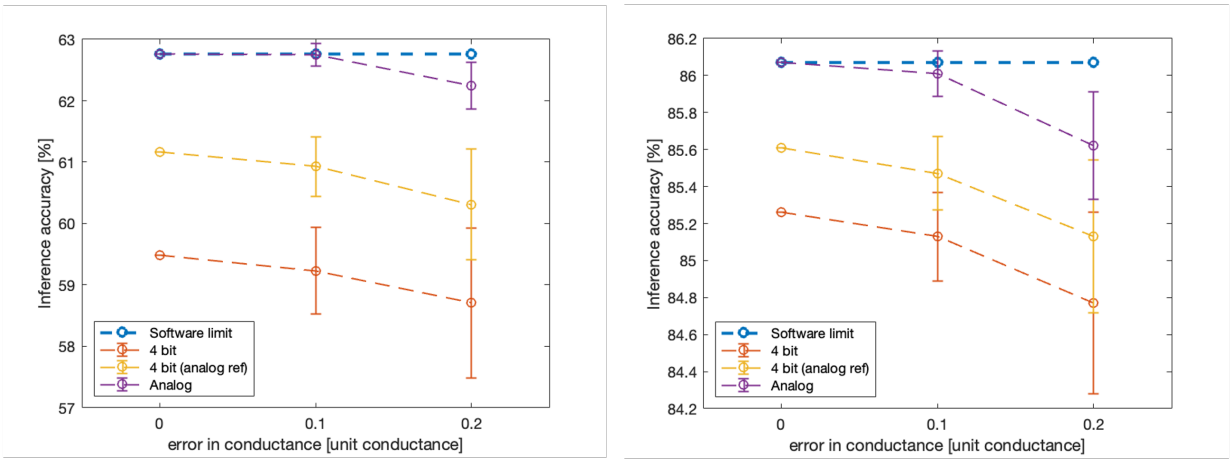


Figure 3.6: The schematic of weight matrix representation of the GPNR array



(a) (b)

Figure 3.7: AI inference simulation with different weight encoding strategies. (a) The inference accuracy for 25-31-31-10 artificial neural networks for 5x5 simplified MNIST handwritten digit classification task. (b) The inference accuracy for 63-63-63-10 artificial neural networks for 5x5 simplified MNIST handwritten digit classification task.

Table 3.1: The AI inference simulation configuration

Cases	Computing method	G_{weight}	G_{bias}	$G_{reference}$
Software limit	Digital	FP32	FP32	FP32
4-bit	Analog	4-bit	4-bit	4-bit
4-bit (analog ref.)	Analog	4-bit	4-bit	Analog
Analog	Analog	Analog	Analog	Analog

As shown in figure 3.7, the fully analog strategy is superior to strategies using quantized GPNR devices, which is understandable since fully analog GPNR devices minimally lose the weight information. However, interestingly, even with twice larger deviation in conductance states in the fully analog strategy than any other strategies, the fully analog strategy still demonstrates its better performance than others. Considering this result, therefore, the fully analog strategy was found as an optimal way to engineer the GPNR device, leaving the question of how to make an infinite number of different geometries.

3.4 GPNR crossbar array fabrication

Based on comprehensive analysis on GPNR device, array and inference perspectives, optimal device geometry parameters, desired channel material conductivity, and conductance allocation on GPNR array has been determined. In this section, device fabrication process and automatic mask drawing software are discussed.

As mentioned in the previous section, the fully analog conductance state strategy for the GPNR array can accomplish minimized AI inference accuracy penalty due to the weight value deviation. However, as the number of the weight in the 32x32 array is already 784, and the state-of-art deep neural network architecture requires more than billions of weights, drawing analog conductance state GPNR channel geometry should be automated. Thus, automatic GPNR geometry and mask drawing software were developed, with which the weight matrix input can be directly converted into the dxf mask file like the example in Fig. 3.8.

The channel material of indium-tin-oxide (ITO) was selected due to its favorable material

conductivity compatible with the 4-8 bit resolution 32x32 GPNR array. To fabricate the GPNR crossbar array, the ITO was sputtered on thermally grown silicon dioxide on top of a silicon wafer. The first Au electrode layer was deposited with an e-beam evaporator with a thin Cr adhesion layer, followed by the lift-off process. Then, the ITO etching with chlorine gas by reactive ion etching (RIE) was performed with certain geometries for each conductance state, followed by wet-etching with buffered oxide etchant for residue removal. Then, the surface was passivated with silicon dioxide by plasma-enhanced chemical vapor

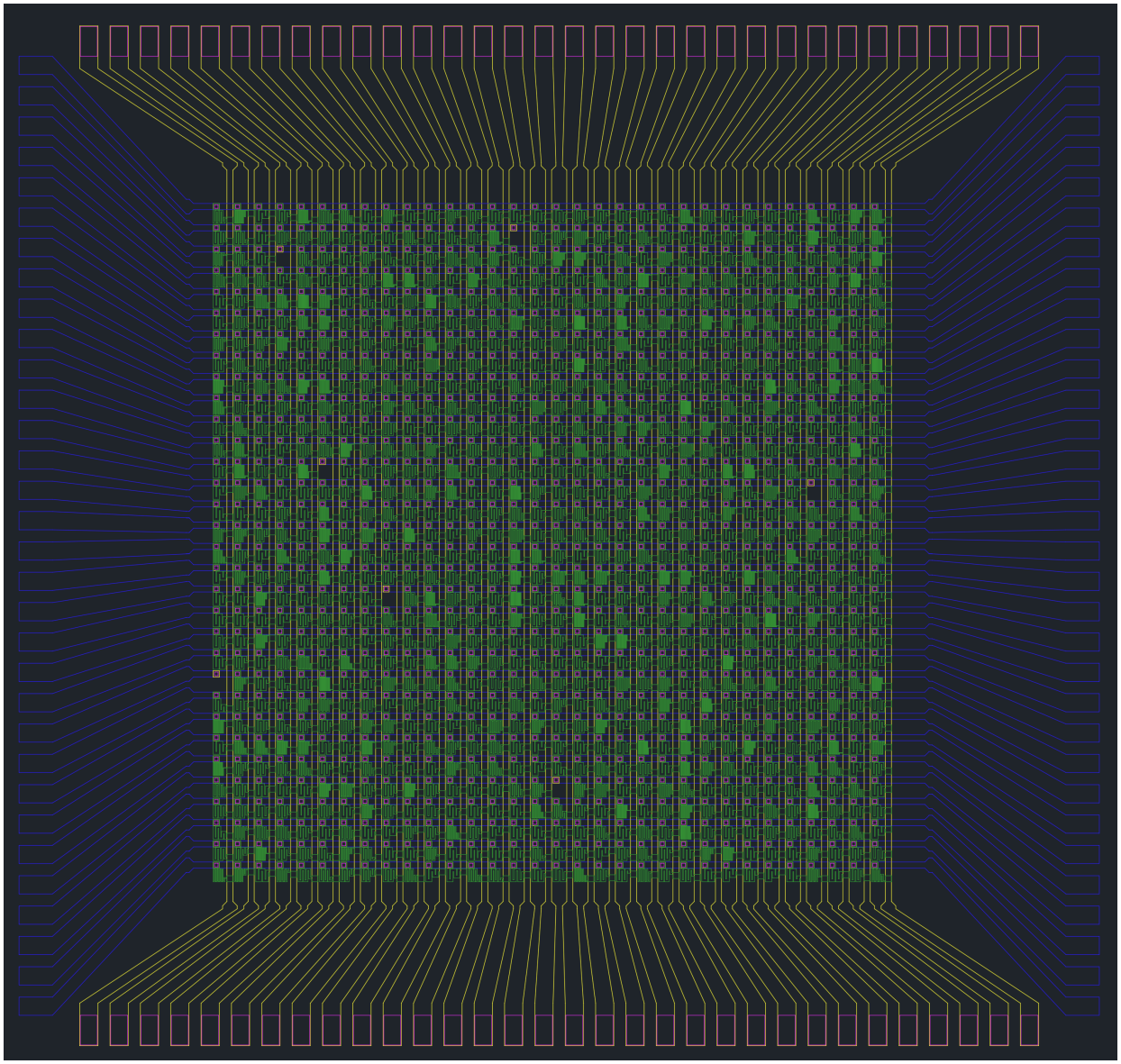


Figure 3.8: Automatically drawn 32x32 GPNR array resistor pattern

deposition (PECVD), and the via holes for the metal pad were etched by RIE. Finally, the second Au electrode layer was sputtered with an adhesion Ti layer, and a lift-off process was conducted to complete the fabrication process. Note that all photolithography process was done by GCA stepper, providing 5x minimized exposure for the maximum lithography resolution on a lab scale.

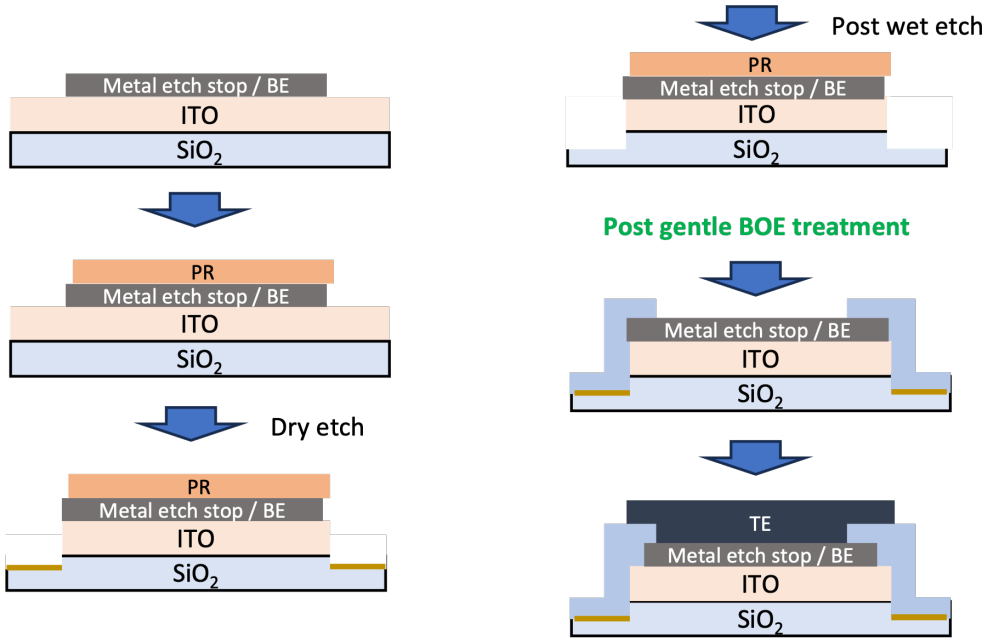


Figure 3.9: ITO channel-based GPNR fabrication process

Chapter 4

GPNR Device Performance and Computing Demonstration

In this chapter, the preliminary results from ITO channel-based GPNR devices shown in Fig. 4.1 are introduced. Overall, the GPNR device exhibits great conductance state accuracy and conductance state stability over time. In addition, the vector-matrix multiplication (VMM) performance was experimentally verified. Finally, the AI inference simulation with a measured GPNR conductance array was performed to evaluate the potential of GPNR devices for AI computing.

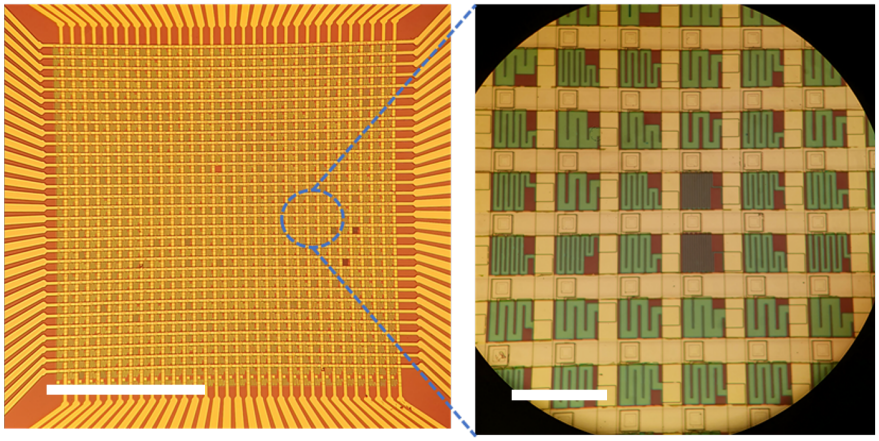
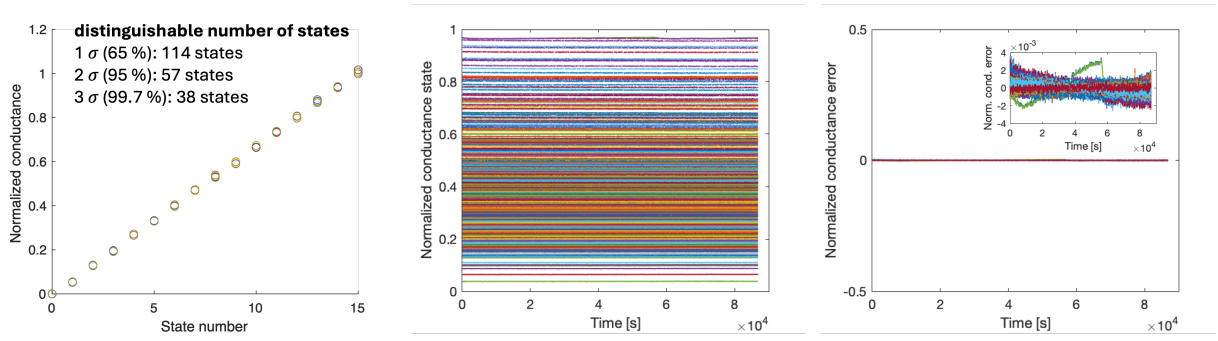


Figure 4.1: Fully-fabricated GPNR device crossbar array. Scalebar lengths are 1 mm and 100 um.

4.1 GPNR device performance

To begin with, the conductance programming accuracy and stability of GPNR devices are first validated (Fig. 4.2). As shown in Fig. 4.2a, the five groups of single GPNR devices with 16 conductance states were fabricated, and they demonstrated 6-bit resolution (114 distinguishable numbers of states, obtained by dividing the total conductance range by twice the standard deviation of conductance error). This programming accuracy is the state-of-the-art performance, compared with most advanced programming performance [24], [26], [27], which makes GPNR device further attractive since current accuracy can be improved with the most advanced photolithography.

On top of conductance programming accuracy, the stability of GPNR devices across time is evaluated. The 28x28 GPNR crossbar array with neural weight matrix was fabricated, and its conductance states were measured for 24 hours (Fig. 4.2b and Fig. 4.2c). It demonstrates 8-bit conductance state stability for 24 hours, which is lower than the state-of-the-art performance done in 1,000 sec ($<$ 1 hour) (11-bit in Ref. [24]), but this is the best performance for the 24-hour experimental demonstration to the best of my knowledge.



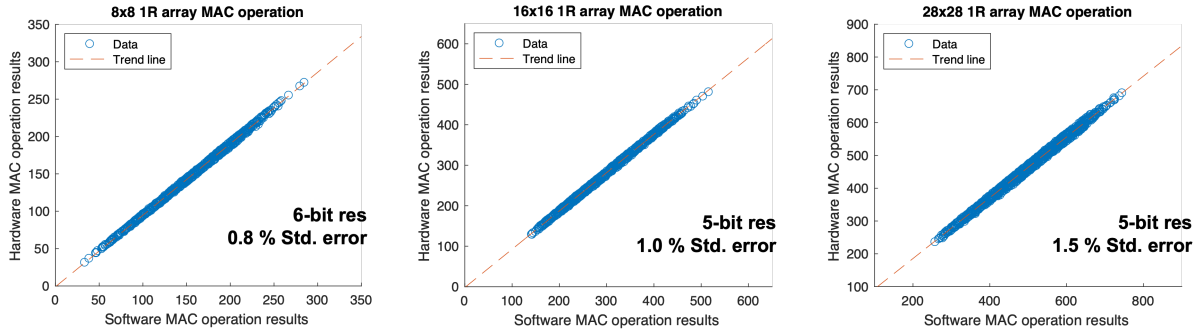
(a) (b) (c)

Figure 4.2: The GPNR device performance. (a) Normalized conductance states of 80 GPNR devices having five groups of 0th-15th conductance states. Conductance states are normalized by twice each group's average conductance value. (b) Normalized conductance states of 784 GPNR devices measured for 24 hours. (c) Normalized conductance error of 784 GPNR devices measured for 24 hours. The inset figure in (c) is drawn with the same data by rescaling the y-axis.

4.2 Vector-matrix multiplication (VMM) performance

Since the state-of-the-art performance of GPNR devices is once checked, the performance of the AI computing operation (vector-matrix multiplication) is experimentally evaluated. Vector-matrix multiplication (VMM) was performed with a field-programmable gate array (FPGA)-based measurement system that can perform VMM operation up to 1 kHz.

As summarized in Fig. 4.3, 1,000 times of VMM operation was performed with randomly generated voltage input ranging from 0 to 1 V. In the smallest array, the best computing accuracy of 6-bit is obtained, while the accuracy gets lower when the array size gets larger. However, the GPNR array used in this demonstration had effective unit conductance of 5.09 µS, 1.27 µS, and 0.318 µS for 4-,6-, and 8-bit, respectively. As a result, the accuracy limit is already restricted up to 5-6 bits considering the equation 3.1, which can be improved with channel materials with a lower conductivity. Nonetheless, this records the best VMM accuracy for a resistive component-based array without selectors while even demonstrating comparable performance to the state-of-the-art NVM arrays with selectors (6-bit for 64x64 array), which prevent issues from non-ideal resistances discussed in the previous chapter, but compromise the computing speed, efficiency, and circuit simplicity [31].



(a) (b) (c)

Figure 4.3: The GPNR device vector-matrix multiplication (VMM) performance. The experimental VMM operation results of (a) 8x8, (b) 16x16, and (c) 28x28 GPNR arrays with randomly generated input vectors ranging from 0 to 1. The VMM outputs are compared with estimated VMM results calculated with a measured conductance array.

4.3 AI inference simulation

Finally, the preliminary AI inference performance of the GPNR crossbar array is checked by simulation with experimentally measured conductance array from GPNR arrays. The neural network is composed of four neural layers with neuron numbers of 25, 31, 31, and 10, resulting in the weight matrix of 25x31, 31x31, and 31x10. The target and fabricated weight matrix are shown in Fig. 4.4, and each weight matrix is fabricated with around 4-, 2-, and 4-bit resolution.

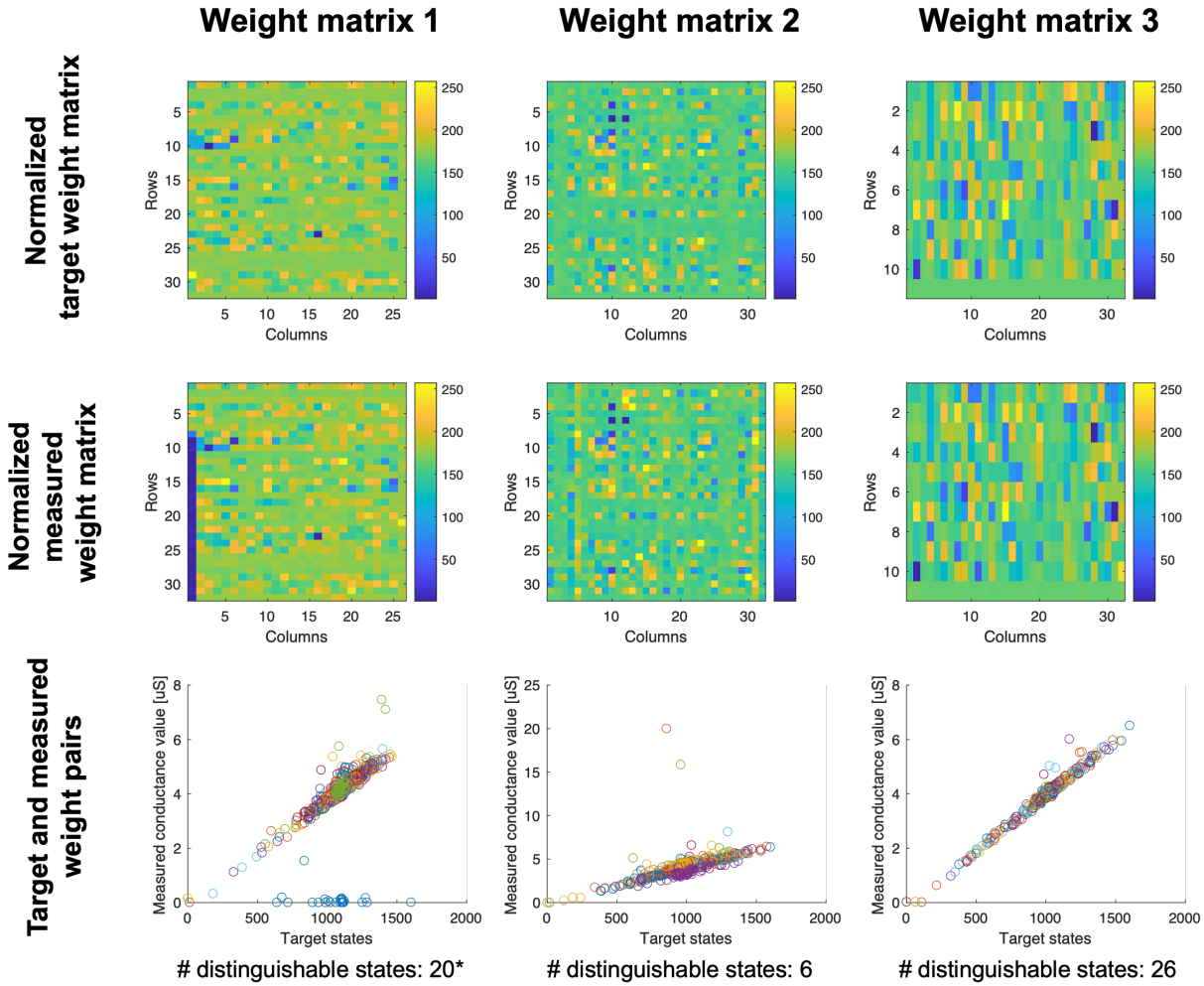


Figure 4.4: Fully-fabricated neural weight matrix. From left to right, the statistics of the first, second, and third neural layers are summarized. * The number of distinguishable of the first layer was calculated without first column values.

Table 4.1: 5x5 simplified MNIST handwritten image classification task inference simulation results

Cases	$W_{1,2}$	$W_{2,3}$	$W_{3,4}$	Accuracy [%]	Exp. accuracy [%]*
Case 1	GPNR	GPNR	GPNR	22.3	18.0 ± 5.4
Case 2	SW	GPNR	GPNR	59.8	66.3 ± 9.0
Case 3	SW	GPNR	SW	67.6	69.3 ± 6.9
Case 4	SW	SW	GPNR	91.0	92.6 ± 0.4
Reference	SW	SW	SW	93.2	NA

* Expected inference accuracy and standard deviation for the weight matrix with the same weight value deviation from GPNR arrays.

The measured conductance matrix was used to perform VMM operation based on test image inputs, and after each of the VMM operations, outputs were activated by the ReLU function while the last ten outputs were compared to each other to estimate the label of the input image. The results are summarized in Table 4.1. When AI inference is performed only with measured GPNR arrays, it gives 22.3 % accuracy, and when the first weight matrix is replaced with the target, it accomplishes 59.8 %. This would imply that the poor fabrication of the first column of the first weight matrix would make the GPNR array lose critical information about the feature, resulting in poor performance. Meantime, when the second layer with poor programming accuracy is further replaced with the target weight matrix, the inference accuracy becomes 91.0 % which has 97.6 % reliability compared to software reference 93.2 %.

Although the accuracy from inference simulation is quite deviated from software reference, the simulation results with artificially generated weight matrix with the weight value deviation in GPNR array are close to the obtained accuracy. This means that if the conductance accuracy in the GPNR array can be improved, the inference accuracy will be enhanced as well. Therefore, higher performance can be readily achieved with the newly carefully made array.

Chapter 5

Conclusion

In conclusion, this thesis has successfully demonstrated the effectiveness of Geometrically Programmed Nano-Resistors (GPNRs) in addressing the critical non-idealities of non-volatile memory (NVM) in analog in-memory computing applications. The research has made significant strides in overcoming the inherent limitations of NVMs by developing a robust and accurate GPNR device that promises enhanced performance in AI acceleration tasks. The experimental results, particularly in AI inference simulations, validate the GPNR's potential to deliver high-accuracy computing operations, affirming its role as a superior alternative to traditional NVM-based systems.

The findings of this work not only pave the way for a more reliable analog AI accelerator but also contribute to making analog AI accelerators suitable for the system-on-a-chip (SoC) system, the most advanced integrated circuit. By minimizing the dependency on high voltage and current programming methods and optimizing the geometric configuration of nano-resistors, the GPNR technology readily fits into the SoC operating condition, which was a challenge in NVM-based AI accelerators. This approach makes GPNR devices particularly valuable for future developments in edge computing and mobile AI applications.

Meanwhile, to further improve this GPNR-based analog AI accelerator attractive to the real application, achieving 8-bit programming accuracy, high yield rate, and 256x256 large

array are needed from a fabrication perspective. In addition, as this GPNR device would compromise the programmability of conventional NVM technology, consideration of adding reconfigurability will significantly enlarge the potential applications. Furthermore, given that this GPNR array can easily achieve low conductance and is the largest selector-less array system, it has a strong potential for several orders of acceleration compared to an NVM-based array system with selectors. To fully implement this, a fast and efficient low-current measurement circuit should be simultaneously investigated.

Looking forward, the implications of this research extend beyond immediate technological advancements. The successful integration of GPNRs into AI systems holds promise for the development of more sustainable, power-efficient computing architectures. Continued refinement and scaling of GPNR technology will be crucial in meeting the growing demands for AI capabilities across various industries. As this technology matures, it will likely catalyze further innovations in the semiconductor sector, driving forward the next generation of AI computing solutions.

References

[1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[4] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*, PMLR, 2019, pp. 6105–6114.

[5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[6] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020.

[7] O. M. Andrychowicz, B. Baker, M. Chociej, R. Jozefowicz, B. McGrew, J. Pachocki, A. Petron, M. Plappert, G. Powell, A. Ray, *et al.*, "Learning dexterous in-hand ma-

nipulation," *The International Journal of Robotics Research*, vol. 39, no. 1, pp. 3–20, 2020.

[8] E. Kaufmann, L. Bauersfeld, A. Loquercio, M. Müller, V. Koltun, and D. Scaramuzza, "Champion-level drone racing using deep reinforcement learning," *Nature*, vol. 620, no. 7976, pp. 982–987, 2023.

[9] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill, *et al.*, "On the opportunities and risks of foundation models," *arXiv preprint arXiv:2108.07258*, 2021.

[10] A. Mehonic and A. J. Kenyon, "Brain-inspired computing needs a master plan," *Nature*, vol. 604, no. 7905, pp. 255–260, 2022.

[11] K. Rupp. "40 years of microprocessor trend data." (2018), URL: https://www.karlrupp.net/2015/06/40-years-of-microprocessor-trend-data/ (visited on 04/17/2024).

[12] A. de Vries, "The growing energy footprint of artificial intelligence," *Joule*, vol. 7, no. 10, pp. 2191–2194, 2023.

[13] A. Sebastian, M. Le Gallo, R. Khaddam-Aljameh, and E. Eleftheriou, "Memory devices and applications for in-memory computing," *Nature nanotechnology*, vol. 15, no. 7, pp. 529–544, 2020.

[14] T. Gokmen and Y. Vlasov, "Acceleration of deep neural network training with resistive cross-point devices: Design considerations," *Frontiers in neuroscience*, vol. 10, p. 203 376, 2016.

[15] M. Lanza, A. Sebastian, W. D. Lu, M. Le Gallo, M.-F. Chang, D. Akinwande, F. M. Puglisi, H. N. Alshareef, M. Liu, and J. B. Roldan, "Memristive technologies for data storage, computation, encryption, and radio-frequency communication," *Science*, vol. 376, no. 6597, eabj9979, 2022.

[16] D. Marković, A. Mizrahi, D. Querlioz, and J. Grollier, "Physics for neuromorphic computing," *Nature Reviews Physics*, vol. 2, no. 9, pp. 499–510, 2020.

[17] Z. Wang, H. Wu, G. W. Burr, C. S. Hwang, K. L. Wang, Q. Xia, and J. J. Yang, "Resistive switching materials for information processing," *Nature Reviews Materials*, vol. 5, no. 3, pp. 173–195, 2020.

[18] D. Ielmini and R. Waser, *Resistive switching: from fundamentals of nanoionic redox processes to memristive device applications*. John Wiley & Sons, 2015.

[19] S. Choi, S. H. Tan, Z. Li, Y. Kim, C. Choi, P.-Y. Chen, H. Yeon, S. Yu, and J. Kim, "Sige epitaxial memory for neuromorphic computing with reproducible high performance based on engineered dislocations," *Nature materials*, vol. 17, no. 4, pp. 335–340, 2018.

[20] H. Yeon, P. Lin, C. Choi, S. H. Tan, Y. Park, D. Lee, J. Lee, F. Xu, B. Gao, H. Wu, *et al.*, "Alloying conducting channels for reliable neuromorphic computing," *Nature nanotechnology*, vol. 15, no. 7, pp. 574–579, 2020.

[21] G. W. Burr, M. J. Brightsky, A. Sebastian, H.-Y. Cheng, J.-Y. Wu, S. Kim, N. E. Sosa, N. Papandreou, H.-L. Lung, H. Pozidis, *et al.*, "Recent progress in phase-change memory technology," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 6, no. 2, pp. 146–162, 2016.

[22] V. Garcia and M. Bibes, "Ferroelectric tunnel junctions for information storage and processing," *Nature communications*, vol. 5, no. 1, p. 4289, 2014.

[23] S. Bhatti, R. Sbiaa, A. Hirohata, H. Ohno, S. Fukami, and S. Piramanayagam, "Spintronics based random access memory: A review," *Materials Today*, vol. 20, no. 9, pp. 530–548, 2017.

[24] M. Rao, H. Tang, J. Wu, W. Song, M. Zhang, W. Yin, Y. Zhuo, F. Kiani, B. Chen, X. Jiang, *et al.*, "Thousands of conductance levels in memristors integrated on cmos," *Nature*, vol. 615, no. 7954, pp. 823–829, 2023.

[25] T.-H. Wen, J.-M. Hung, W.-H. Huang, *et al.*, "Fusion of memristor and digital compute-in-memory processing for energy-efficient edge computing," *Science*, vol. 384, no. 6693, pp. 325–332, 2024. DOI: 10.1126/science.adf5538.

[26] W. Song, M. Rao, Y. Li, C. Li, Y. Zhuo, F. Cai, M. Wu, W. Yin, Z. Li, Q. Wei, *et al.*, "Programming memristor arrays with arbitrarily high precision for analog computing," *Science*, vol. 383, no. 6685, pp. 903–910, 2024.

[27] W. Wan, R. Kubendran, C. Schaefer, S. B. Eryilmaz, W. Zhang, D. Wu, S. Deiss, P. Raina, H. Qian, B. Gao, *et al.*, "A compute-in-memory chip based on resistive random-access memory," *Nature*, vol. 608, no. 7923, pp. 504–512, 2022.

[28] M. Le Gallo, R. Khaddam-Aljameh, M. Stanisavljevic, A. Vasilopoulos, B. Kersting, M. Dazzi, G. Karunaratne, M. Brändli, A. Singh, S. M. Mueller, *et al.*, "A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference," *Nature Electronics*, vol. 6, no. 9, pp. 680–693, 2023.

[29] S. Ambrogio, P. Narayanan, A. Okazaki, A. Fasoli, C. Mackin, K. Hosokawa, A. Nomura, T. Yasuda, A. Chen, A. Friz, *et al.*, "An analog-ai chip for energy-efficient speech recognition and transcription," *Nature*, vol. 620, no. 7975, pp. 768–775, 2023.

[30] M. Le Gallo, D. Krebs, F. Zipoli, M. Salinga, and A. Sebastian, "Collective structural relaxation in phase-change memory devices," *Advanced Electronic Materials*, vol. 4, no. 9, p. 1 700 627, 2018.

[31] C. Li, M. Hu, Y. Li, H. Jiang, N. Ge, E. Montgomery, J. Zhang, W. Song, N. Dávila, C. E. Graves, *et al.*, "Analogue signal and image processing with large memristor crossbars," *Nature electronics*, vol. 1, no. 1, pp. 52–59, 2018.